Discrete Mathematics

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Lectures: Tuesday 8-9, Thursday 8-9

Preface

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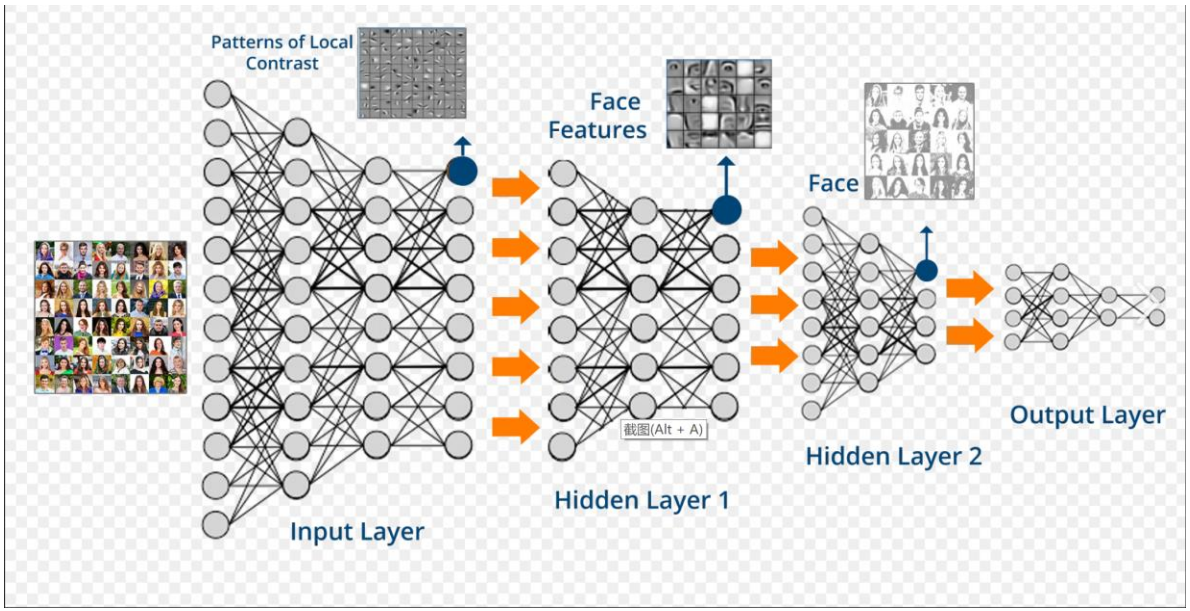
Continuous vs Discrete

光滑连续（微分、积分） VS 离散、有限、近似

实际问题---离散表示---数学建模---解决问题

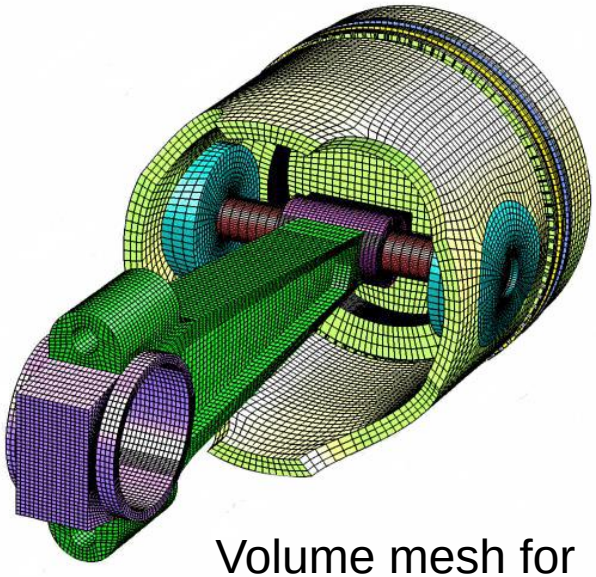


Pixels of image

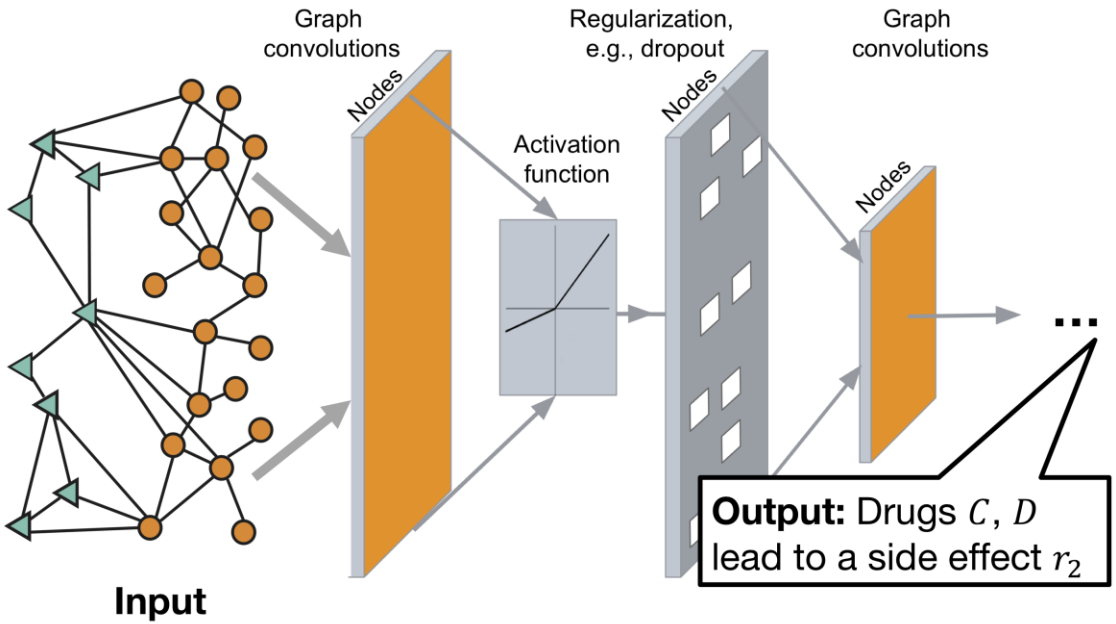


Convolutional Neural network

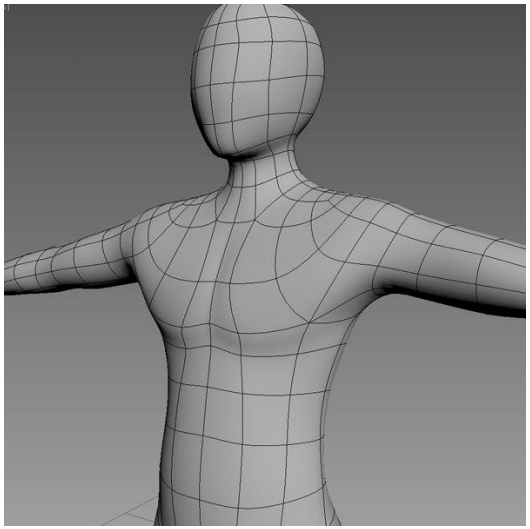
Continuous vs Discrete



Volume mesh for industrial simulation

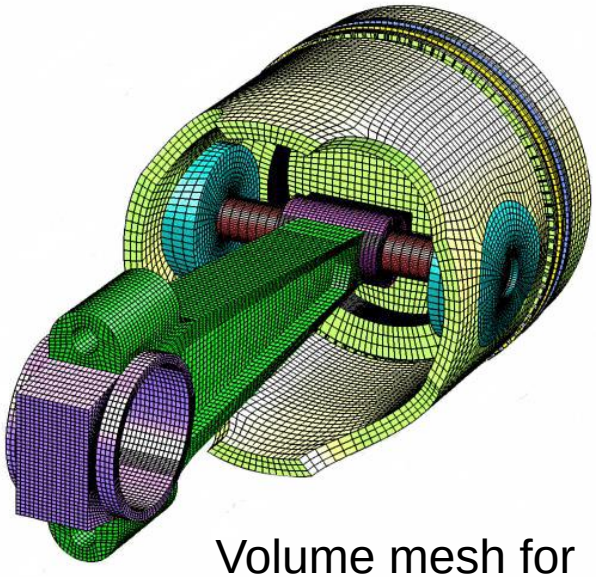


Graph Neural Network

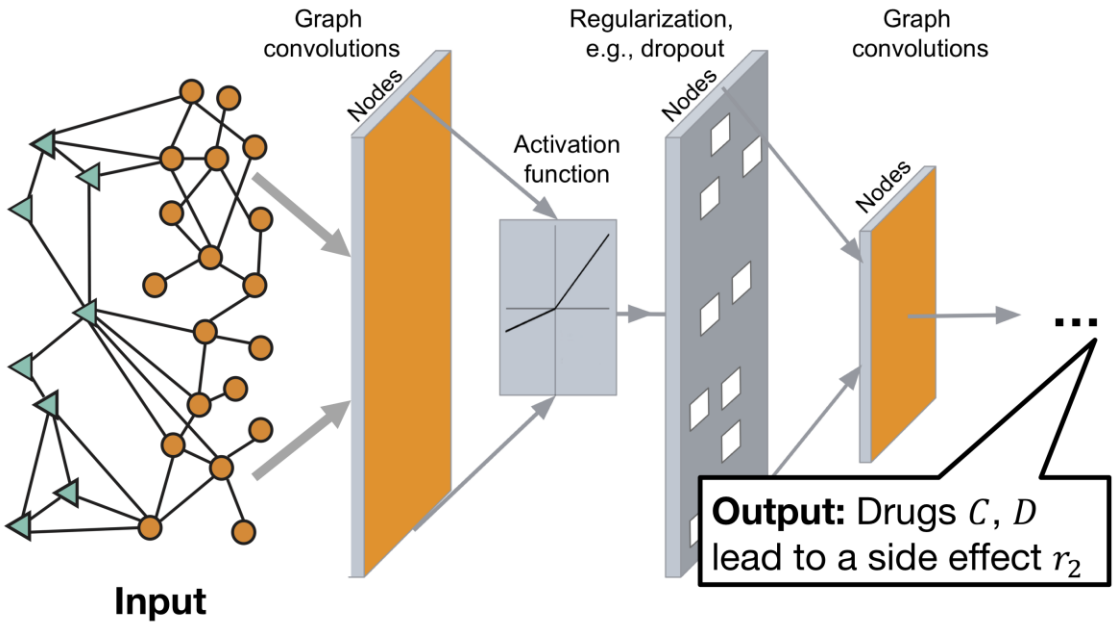


Mesh model for animation

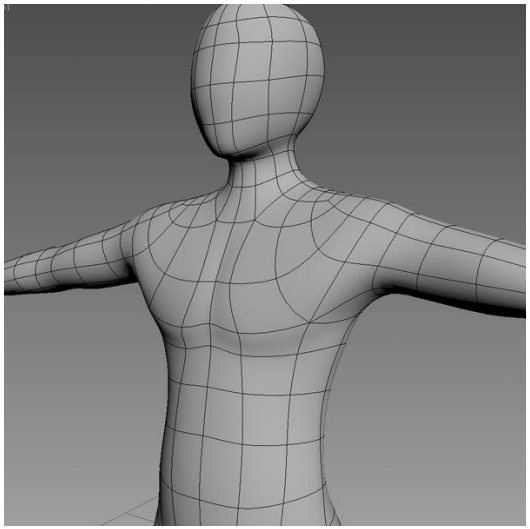
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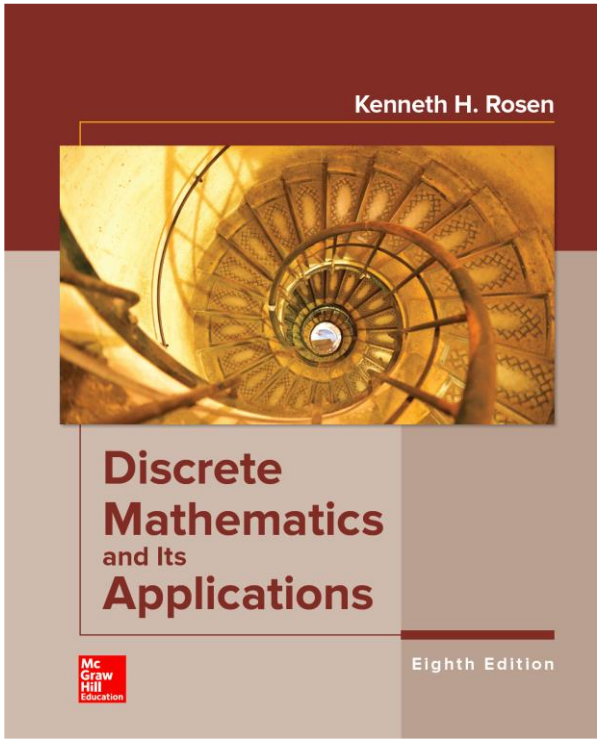


Mesh model for animation

Course Organization

Textbook: Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, Eighth Edition, McGraw-Hill, 2019

http://www.mhhe.com/rosen



Course Organization

Assignments: There will be an assignment every week. Each student is expected to solve the assignments independently, and submits a solution for every assigned problem.

Examinations: The final examination will be a two-hour examination at the end of the term. To pass the course or not is dependent on:

Final examination: = 50%

Midterm examination = 25%

Quiz and Assignments = 25%

Goal of the Course [1]

- How to think mathematically
  - Mathematically Reasoning: Logic
  - Combinational Analysis: Counting
  - Discrete Structure: sets, Permutation, Relations, Graphs, Trees, Finite state machine
  - Algorithm Thinking: Computer Program (memory and time complex)

Goal of the Course [2]

- Applications and modeling
  - Computer Science: Binary number, Finite memory
    - How many ways are there to choose a valid password on a computer system?
    - How can a circuit that adds two integers be designed?
  - Other applications: Communication, Automatic Control
    - Is there a link between two computers in a network?
    - How to solve a discrete optimization problem?
- Gateway to more advanced courses
  - Data Structure, Database, Operating Systems, Compiler Theory, AI

Framework of the Course

- Set
- Logic & Reasoning
- Counting
- Relations & Graph Theory
- Algebra System

Study discrete objects and their relations

The Foundations: Logic and Proofs

1

Chapter 1

- 1.1-1.2 Propositional Logic
- 1.3 Propositional Equivalences
- 1.4 Predicates and Quantifiers
- 1.5 Nested Quantifiers
- 1.6 Rules of Inference
- 1.7 Introduction to Proofs
- 1.8 Proof Methods and Strategy

Mathematical Logic

- Traditional logic as a part of philosophy is one of the oldest scientific disciplines. It can be traced back to the Stoics and to Aristotle and is the root of what is nowadays called philosophical logic.
- Mathematical logic—evolving around the notions of logical validity, provability, and computation was created in the first half of the previous century by a cohort of brilliant mathematicians and philosophers such as Frege, Hilbert, Godel, Turing, Tarski, Malcev, Gentzen, and some others. The development of this expanded mathematics into a novel area of applications, subjected logical reasoning and computability to rigorous analysis, and eventually led to the creation of computers.

Mathematical Logic

- Mathematical logic: specify the precise meaning of mathematical statements
- One feature of modern logic is a clear distinction between object language and metalanguage.
- It should be noticed that logic used in the sciences differs essentially from logic used in everyday language, where logic is more an art than a serious task of saying what follows from what.
- The way of arguing about formal languages and theories is traditionally called the *metatheory*. An important task of *a metatheoretic analysis is* to specify procedures of logical inference by so-called *logical calculi, which* operate purely syntactically.

Additional Reading

- The relatively short period of development of modern mathematical logic in the twentieth century.
- See Paolo Mancosu, Richard Zach, Calixto Bade "The Development of Mathematical Logic from Russell to Tarski: 1900–1935"

1.1 Propositional Logic

【**Definition**】Proposition is a statements that is either true or false, but not both. (contrary to Fuzzy Logic)

- Examples
  - What time is it ? ×
  - Read this carefully. ×
  - X+1=2 ×
  - Toronto is the capital of Canada. √
  - 1+1=2 √
- Conventional letters: denote propositions.

  Such as *p, q, r, s* …
- True values: T (true proposition), F (false proposition)
- Propositional logic (calculus): the area of logic that deals with proposition.

Compound proposition

- Compound propositions are formed from existing propositions using logical operators.
- We formalize this by denoting propositions with letters such as p, q, r, s, and introducing several logical operators (Connectives).

- Logical Operations
  - negation operator (NOT)
  - conjunction operator (AND)
  - disjunction operator (OR)
  - exclusive or operator (XOR)
  - implication operator (if-then)
  - biconditional operator (If and only if)
- True table: displays the relationships between the true values of propositions.

Negation (NOT)

- p: I am going to town.
- $\neg p$: I am not going to town.
- Truth table:

p	$\neg p$
T	F
F	T

Conjunction (AND)

- p: I am going to town.
- q: It is raining.
- $p \wedge q$: I am going to town and it is going to rain.
- Both p and q must be true for the conjunction to be true.
- Truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction (OR)

- p: I am going to town.
- q: It is raining.
- $p \vee q$: I am going to town or it is raining.
- Inclusive or - only one proposition needs to be true for the disjunction to be true.
- Truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example

- Consider the following propositions:

  p: Today is Monday.

  q: I will go to the beach.

  Then,

  $p \wedge q$: Today is Monday and I will go to the beach.

  $p \vee q$: Today is Monday or I will go to the beach.

  $\neg p \wedge q$: Today is not Monday and I will go to the beach.

  $\neg p \wedge \neg q$: Today is not Monday and I will not go to the beach.

Exclusive OR (XOR)

- p: I am going to town.
- q: It is raining.
- $p \oplus q$: Either I am going to town or it is raining.
- Only one of p and q are true. (not both)
- Truth table:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication - If .. then

- p: I am going to town.
- q: It is raining.
- $p \rightarrow q$: If I am going to town, then it is raining.
- The implication is false only when p is true and q is false!
- Truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Implication - Equivalent Forms

- If *p*, then *q*
- *p* implies *q*
- If *p*, *q*
- *q* if *p*
- *q* whenever *p*
- *p* is a sufficient condition for *q*
- *q* is a necessary condition for *p*
- *p* only if *q*

Converse of an Implication

- Implication: $p \rightarrow q$
- Converse: $q \rightarrow p$
- Implication:
  - If I am going to town, it is raining.
- Converse:
  - If it is raining, then I am going to town.

Contrapositive of an Implication

- Implication: $p \rightarrow q$
- Contrapositive: $\neg q \rightarrow \neg p$
- Implication:
  - If I am going to town, it is raining.
- Contrapositive:
  - If it is not raining, then I am not going to town.

When two compound propositions always have the same truth value we call them *equivalent*.

Biconditional

- "if and only if", "iff"
- $p \leftrightarrow q$
- I am going to town if and only if it is raining.
- Both *p* and *q* must have the same truth value for the assertion to be true.
- Truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Truth Table Summary

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	F	T	T
T	F	F	T	T	F	F
F	T	F	T	T	T	F
F	F	F	F	F	T	T

Constructing a Truth Table

- One column for each propositional variable
- One column for each compound proposition
- For n propositional variables there will be 2^n rows

Example

- $(p \vee q) \rightarrow \neg r$

p	q	r	$p \vee q$	$\neg r$	$(p \vee q) \rightarrow \neg r$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	F	T	T

Precedence of Logical Operators

- Parentheses gets the highest precedence
- Then $\neg \quad \wedge \quad \vee \quad \rightarrow \quad \leftrightarrow$
- Example:

  $p \wedge q \vee r$ means $(p \wedge q) \vee r$, not $p \wedge (q \vee r)$

  $p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$

Logic and Bit Operations [1]

- *Bit:* has two possible values , namely, 0 and 1
- A *Boolean variable* is one whose value is either true or false.
- A Boolean variable can be represented using a bit.
- Computer *bit operations* correspond to logical operations of Boolean variables.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Logic and Bit Operations [2]

- A *bit string* is a sequence of zero or more bits.
- The *length* of this string is the number of bits in the string.
- *Bitwise operations* are bit operations extended to bit strings.
- The bitwise of two strings of the same length:
  - Bitwise OR
  - Bitwise AND
  - Bitwise XOR

Example

Find the bitwise OR, bitwise AND, and bitwise XOR of the bit strings 01 1011 0110 and 11 0001 1101.

01 1011 0110	
11 0001 1101	
11 1011 1111	bitwise OR
01 0001 0100	bitwise AND
10 1010 1011	birwise XOR

1.2 Applications of Propositional Logic

- Translating English Sentences
- System Specifications
- Boolean Searches
- Logic Puzzles
- Logic Circuits

Translating English sentences

- Informal languages are changed to formal/mathematical languages.

- Example 12 (p.11): How can the following sentence be translated into a logical expression?

  "*You can access the Internet from campus only if you are a computer science major or you are not a freshman.*"

Solutions:

Let a, c and f represent "you can access the Internet from campus","you are a computer science major" and "you are a freshman".

This sentence can be represented as

$$a \rightarrow (c \vee \neg f)$$

Self-study

- System Specifications
- Boolean Searches
- Logic Circuits

Logic Puzzles

- An island has two kinds of inhabitants, *knights*, who always tell the truth, and *knaves*, who always lie.
- You go to the island and meet A and B.
  - A says “B is a knight.”
  - B says “The two of us are of opposite types.”

Example: What are the types of A and B?

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively. So, then $\neg p$ represents the proposition that A is a knave and $\neg q$ that B is a knave.

- If A is a knight, then p is true. Since knights tell the truth, q must also be true. Then $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true, but it is not. So, A is not a knight and therefore $\neg p$ must be true.
- If A is a knave, then B must not be a knight since knaves always lie. So, then both $\neg p$ and $\neg q$ hold since both are knaves.

Exercise Problems

- P.13: 13, 15, 18, 30, 36, 44, 48
- P.23: 5, 8, 23-27, 28-35, 40, 44

1.3 Propositional Equivalences

- Basic Terminology
- Logical Equivalences
- Satisfiability

Basic Terminology

- A *tautology* is a proposition which is always true.

$$p \vee \neg p$$

- A *contradiction* is a proposition that is always false.

$$p \wedge \neg p$$

- A *contingency* is a proposition that is neither a tautology nor a contradiction.

$$p \vee q \rightarrow \neg r$$

$$(p \vee \neg q) \wedge (q \vee \neg p) \quad ??$$

Logical Equivalences

【**Definition**】Two propositions p and q are logically equivalent if they have the same truth values in all possible cases.

- Two propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- Notation: $p \Leftrightarrow q$ or $p \equiv q$

Determining Logical Equivalence

Using a truth table

- Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg (p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

- Not a very efficient method, WHY?
- Solution: Develop a series of equivalences.

Logical Equivalence and Comprehension

- Truth tables for $\neg p \vee q \Leftrightarrow p \rightarrow q$

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	F	T	T

p: You don't work hard;

q: You will be fired

$\neg p \vee q$: You must work hard or you will be fired

$p \rightarrow q$: If you don't work hard then you will be fired

Important Equivalences [1]

$p \wedge T \Leftrightarrow p$	Identity（同一）
$p \vee F \Leftrightarrow p$	
$p \vee T \Leftrightarrow T$	Domination（支配）
$p \wedge F \Leftrightarrow F$	
$p \vee p \Leftrightarrow p$	Idempotent（幂等）
$p \wedge p \Leftrightarrow p$	
$\neg(\neg p) \Leftrightarrow p$	Double Negation

Important Equivalences [2]

$p \vee q \Leftrightarrow q \vee p$	Commutative
$p \wedge q \Leftrightarrow q \wedge p$	
$(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$	Associative
$(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$	
$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$	Distributive
$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$	
$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$	De Morgan's
$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$	

Important Equivalences [3]

$p \vee (p \wedge q) \Leftrightarrow p$	Absorption
$p \wedge (p \vee q) \Leftrightarrow p$	
$p \vee \neg p \Leftrightarrow \mathrm{T}$	Negation
$p \wedge \neg p \Leftrightarrow \mathrm{F}$	

All of these Equivalences can be proved by Truth table. We can use them freely without proof any more.

Using De Morgan's Laws

- Example: Use De Morgan's laws to express the negation of "Miguel has a cellphone and he has a laptop computer" and "Heather will go to the concert or Steve will go to the concert".

Solution:

Miguel does not have a cellphone or he does not have a laptop computer.

Heather will not go to the concert and Steve will not go to the concert.

Example [1]

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

$\neg(p \vee (\neg p \wedge q)) \Leftrightarrow \neg p \wedge \neg(\neg p \wedge q)$ DeMorgan's 2nd law

$\Leftrightarrow \neg p \wedge [\neg(\neg p) \vee \neg q]$ DeMorgan's 1st law

$\Leftrightarrow \neg p \wedge (p \vee \neg q)$ Double negation

$\Leftrightarrow (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ Distributive law

$\Leftrightarrow \mathrm{F} \vee (\neg p \wedge \neg q)$

$\Leftrightarrow (\neg p \wedge \neg q)$ Identity law

Example [2]

- **Show that ((p →q) →r) →((r →p) →(s →p)) is a tautology.**

  Solution:

$$((p \to q) \to r) \to ((r \to p) \to (s \to p))$$

$$\Leftrightarrow \neg(\neg(\neg p \vee q) \vee r) \vee (\neg(\neg r \vee p) \vee (\neg s \vee p))$$

$$\Leftrightarrow ((\neg p \vee q) \wedge \neg r) \vee (r \wedge \neg p) \vee (\neg s \vee p)$$

$$\Leftrightarrow (\neg p \wedge \neg r) \vee (q \wedge \neg r) \vee (r \wedge \neg p) \vee (\neg s \vee p)$$

$$\Leftrightarrow (\neg p \wedge (\neg r \vee r)) \vee (q \wedge \neg r) \vee (\neg s \vee p)$$

$$\Leftrightarrow \neg p \vee (q \wedge \neg r) \vee (\neg s \vee p)$$

$$\Leftrightarrow T$$

Propositional Satisfiability

- A compound proposition is *satisfiable* if there is an assignment of truth values to its variables that make it true. When no such assignments exist, the compound proposition is *unsatisfiable*.
- A compound proposition is unsatisfiable if and only if its negation is a tautology.

Questions on Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

Solution: Satisfiable. Assign T to *p*, *q*, and *r*.

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Satisfiable. Assign T to *p* and *F* to *q*.

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Not satisfiable. Check each possible assignment of truth values to the propositional variables and none will make the proposition true.

Self-study

Applications of satisfiability

Exercises

- P.38: 9, 18, 27, 40-42, 59, 61, 63, 67

1.4 Predicates and Quantifiers

- Predicates
- Quantifiers
- Quantifiers with restricted domains
- Binding variables
- Logical Equivalences Involving Quantifiers
- Negating Quantified Expressions
- Translating from English into Logical Expressions
- Examples from Lewis Carroll
- Using Quantifiers in System Specifications
- Logic Programming

Predicates

- Statements involving variables: a variable x is the subject of the statement, and P is the predicate. $P(x)$ is the value of the propositional function at x. $P(x_1, x_2, \ldots, x_n)$ is the value of the prepositional function P at the n-tuple $(x_1, x_2, \ldots, x_n)$.
- $P(x)$: $x > 3$
  - Variable x is subject of the statement, “is greater than 3” is predicate.
  - P(4)=T, P(2)=F
- $P(x)$: if $x > 0$, then $x := x+1$
  - If $P(x)$ is true, the assignment statement $x := x+1$ is executed. If $P(x)$ is false, the assignment statement $x := x+1$ is not executed.
- x is between y and z. $B(x, y, z)$

Notes

- Propositional function has not a definite truth value.
- Once a value has been assigned to the variable x, $P(x)$ becomes a proposition and has a truth value.
- The *truth value of* $P(x)$ can be determined when x is assigned a value. (The variable x is bound.)
- Example: Let $P(x)$ denote the statement "$x > 0$."
  - Is $P(y) \wedge \neg P(0)$ a proposition?
  - Is $P(3) \vee \neg P(0)$ a proposition?

In general, a statement of the form $P(x_1, x_2, \cdots, x_n)$ is the value of the propositional function *P* at the *n*-tuple $(x_1, x_2, \cdots, x_n)$, and *P* is also called a *n-place predicate* or *a n-ary predicate*.

Quantifiers

- Create a proposition from a propositional function without assigning values to variables.
- The universal quantification of $P(x)$ is the proposition "$P(x)$ is true for all values x of in the universe of discourse." Denoted by $\forall x\, P(x)$.
- The existential quantification of $P(x)$ is the proposition "There exists an element x in the universe of discourse such that $P(x)$ is true." Denoted by $\exists x\, P(x)$.
- When all of the elements in the universe of discourse can be listed: $x_1, x_2, \ldots, x_n$
  - $\forall x\, P(x) \Leftrightarrow P(x_1) \wedge P(x_2) \wedge \ldots \wedge P(x_n)$
  - $\exists x\, P(x) \Leftrightarrow P(x_1) \vee P(x_2) \vee \ldots \vee P(x_n)$

Examples

- "Every student in this class has studied calculus"

  $P(x)$: "x has studied calculus."

  $S(x)$: "x is in this class."

  $\forall x\,(S(x) \rightarrow P(x))$

- $P(x)$: "x < 2", where the universe of discourse is the set of real numbers

  $\forall x\,P(x)$ = F

  $\exists x\,P(x)$ = T

Conclusion

statement	when True?	When False?
$\forall x\, P(x)$	$P(x)$ is true for every x	There is an x for which $P(x)$ is false
$\exists x\, P(x)$	There is an x for which $P(x)$ is true	$P(x)$ is false for every x

Other Quantifiers

- Uniqueness quantifier: $\exists! \, xP(x)$

  There exists a unique x such that $P(x)$ is true

Quantifiers with Restricted Domains

- $\forall x<0(x^2>0)$
- $\forall y\neq 0(y^3\neq 0)$
- $\exists z>0(z^2=2)$

The domain consists of the real number.

Precedence of Quantifiers

- The quantifiers ∃ and ∀ have higher precedence than all logical operators from propositional calculus.

Binding Variables

- Bound: when a quantifier is used on the variable or when we assign a value to this variable, this variable is bound, otherwise this variable is free.
- All the variables that occur in a propositional function must be bound to turn it into a proposition. This can be done using a combination of universal quantifiers, existential quantifiers, and value assignments.
- The order of the quantifiers is important for the mathematical statements involving multiple quantifications of propositional functions with more than one variable, unless all the quantifiers are universal quantifiers or all are existential quantifiers.

Logical Equivalence Involving Quantifiers

- Definition: Statements involving predicates and quantifiers are logically equivalent iff they have the same truth value no matter which predicates are substitute into these statements and which domain of discourse is used for the variables in these propositional function. S≡T

- Show that $\forall x(P(x) \wedge Q(x)) \equiv \forall xP(x) \wedge \forall xQ(x)$

  Page 49 Example 19

Examples

$$\forall x(A(x) \wedge B(x)) \equiv \forall x A(x) \wedge \forall x B(x)$$
$$\exists x(A(x) \vee B(x)) \equiv \exists x A(x) \vee \exists x B(x)$$

Note:

$$\forall x(A(x) \vee B(x)) \not\Leftrightarrow \forall x A(x) \vee \forall x B(x)$$
$$\exists x(A(x) \wedge B(x)) \not\Leftrightarrow \exists x A(x) \wedge \exists x B(x)$$

$$\exists x(A(x) \wedge B(x)) \Rightarrow \exists x A(x) \wedge \exists x B(x)$$
$$\forall x A(x) \vee \forall x B(x) \Rightarrow \forall x(A(x) \vee B(x))$$

Negating Quantified Expression

Negation	Equivalent Statement	When is negation true?	when false?
$\neg\ \exists x\ P(x)$	$\forall x\ \neg P(x)$	$P(x)$ is false for every x	There is an x for which $P(x)$ is true
$\neg\ \forall x\ P(x)$	$\exists x\ \neg P(x)$	There is an x for which $P(x)$ is false	$P(x)$ is true for every

De Morgan's laws for quantifiers

$$\neg\forall x A(x) \equiv \exists x \neg A(x)$$
$$\neg\exists x A(x) \equiv \forall x \neg A(x)$$

Some other logical Equivalences

x is not occurring in P and B.

(1) $\forall x A(x) \vee P \equiv \forall x(A(x) \vee P)$

(2) $\forall x A(x) \wedge P \equiv \forall x(A(x) \wedge P)$

(3) $\exists x A(x) \vee P \equiv \exists x(A(x) \vee P)$

(4) $\exists x A(x) \wedge P \equiv \exists x(A(x) \wedge P)$

(5) $\forall x(B \rightarrow A(x)) \equiv B \rightarrow \forall x A(x)$

Proof:

$$\forall x(B \rightarrow A(x)) \equiv \forall x(\neg B \vee A(x))$$

$$\equiv \neg B \vee \forall x A(x)$$

$$\equiv B \rightarrow \forall x A(x)$$

Translating from English into Logical Expressions

- Goal: To produce a logical expression that is simple and can be easily used in subsequent reasoning.
- Steps:
  - Clearly identify the appropriate quantifier(s)
  - Introduce variable(s) and predicate(s)
  - Translate using quantifiers, predicates, and logical operators

Example

- $C(x)$: x is a CS student, $E(x)$: x is an CM student, $S(x)$: x is a smart student, U = {all students in our class}

  1) Everyone is a CS student.

  $$\forall x\, C(x)$$

  2) Nobody is an CM student.

  $$\forall x \neg E(x) \quad \textbf{or} \quad \neg \exists x\, E(x)$$

  3) All CS students are smart students.

  $$\forall x\, (C(x) \rightarrow S(x))$$

  4) Some CS students are smart students.

  $$\exists x\, (C(x) \wedge S(x))$$

Example

- $C(x)$: x is a CS student, $E(x)$: x is an EE student, $S(x)$: x is a smart student, U = {all students in our class}

5) No CS student is an EE student.

- If x is a CS student, then that student is not an EE student.

$$\forall x\,(C(x) \rightarrow \neg E(x))$$

- There does not exist a CS student who is also an EE student.

$$\neg \exists x\,[C(x) \wedge E(x)]$$

6) If any EE student is a smart student then he is also a CS student.

$$\forall x\,((E(x) \wedge S(x)) \rightarrow C(x))$$

Examples from Lewis Carroll [1]

- "All lions are fierce"

  "Some lions do not drink coffee"

  "Some fierce creatures do not drink coffee"

  Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a lion", "x is fierce", and "x drinks coffee", respectively.

$$\forall x\,(P(x) \rightarrow Q(x))$$
$$\exists x\,(P(x) \wedge \neg R(x))$$
$$\exists x\,(Q(x) \wedge \neg R(x))$$

The second statement can not be written as $\exists x\,(P(x) \rightarrow \neg R(x))$

The reason is that $P(x) \rightarrow \neg R(x)$ is true whenever x is not a lion, so that $\exists x\,(P(x) \rightarrow \neg R(x))$ is true as long as there is at least one creature is not a lion, even if every lion drinks coffee.

Examples from Lewis Carroll [2]

- “All hummingbirds are richly colored”

  “No large birds live on honey”

  “Birds that do not live on honey are dull in color”

  “Hummingbirds are small”

  Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements “x is a hummingbird”, “x is large”, “x lives on honey”, and “x is richly colored”, respectively.

  $$\forall x\,(P(x) \rightarrow S(x))$$
  $$\neg\exists x\,(Q(x) \wedge R(x))$$
  $$\forall x\,(\neg R(x) \rightarrow \neg S(x))$$
  $$\forall x\,(P(x) \rightarrow \neg Q(x))$$

Exercises

- P.56: 15, 20, 25, 33, 34, 43, 52-53, 62-63

1.5 Nested Quantifiers

- Two quantifies are nested if one is within the scope of the other.
- For example, $\forall x \exists y\ (x+y=0)$

Example

- Let $Q(x,y)$ denote "x+y=0". What are the true values of the quantifications $\exists y \forall x\, Q(x,y)$ and $\forall x \exists y\, Q(x,y)$

"There is a real number y such that for every real number x, $Q(x,y)$ is true"

So $\exists y \forall x\, Q(x,y)$ is false

"For every real number x there is a real number y such that $Q(x,y)$ is true"

So $\forall x \exists y\, Q(x,y)$ is true

Summary

Statement	When True?	When False?
$\forall x \forall y\ P(x,y)$ $\forall y \forall x\ P(x,y)$	$P(x,y)$ is true for every pair x, y.	There is a pair x,y for Which $P(x,y)$ is false
$\forall x \exists y\ P(x,y)$	For every x there is a y for which $P(x,y)$ is true	There is an x such that $P(x,y)$ is false for every y
$\exists x \forall y\ P(x,y)$	There is an x for which $P(x,y)$ is true for every y	For every x there is a y for which $P(x,y)$ is false
$\exists x \exists y\ P(x,y)$	There is a pair x, y for which $P(x,y)$ is true	$P(x,y)$ is false for every pair x, y

Translating Mathematical Statements into Statements Involving Nested Quantifiers

- Express the definition of a limit using quantifiers.

Solution:

Recall the definition of the statement $\lim_{x \to a} f(x) = L$

is: For every real number ε>0 there exists a real number δ>0 such that $0 < |x - a| < \delta$

$$\forall \epsilon \exists \delta \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon),$$

$$\forall \epsilon > 0 \; \exists \delta > 0 \; \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

Translating from Nested Quantifiers into English

- Translate the statement $\forall x\,(C(x) \vee \exists y\,(C(y) \wedge F(x,y)))$ into English, where $C(x)$ is "x has a computer", $F(x,y)$ is "x and y are friends", and the domain for both x and y is the set of all students in your school.

  For every student x in your school x has a computer or there is a student y such that y has a computer and x and y are friends.

  Every student in your school has a computer or has a friend who has a computer

Translating from Nested Quantifiers into English

- Translate the statement

$$\exists x \forall y \forall z (((F(x,y) \wedge F(x,z) \wedge (y \neq z)) \rightarrow \neg F(y,z)))$$

into English, where $F(a,b)$ means a and b are friends and the universe of discourse for x, y, and z is the set of all students in your school.

There is a student x such that for all students y and all students z other than y, if x and y are friends and x and z are friends, then y and z are not friends.

There is a student none of whose friends are also friends with each other.

Translating English Sentences into Logical Expressions

- Express the statement “Everyone has exactly one best friend” as a logical expression.

  Let $B(x,y)$ be the statement “y is the best of x”

  $\forall x \exists y \forall z (B(x,y) \wedge ((z \neq y) \rightarrow \neg B(x,z)))$

- Use quantifiers to express the statement “There is a woman who has taken a flight on every airline in the world”.

  Let $P(w,f)$ be “w has taken f ” and $Q(f,a)$ be “f is a flight on a”.

  $\exists w \forall a \exists f (P(w,f) \wedge Q(f,a))$

Translating English Sentences into Logical Expressions

- Express the statement "All humans have some shortcoming" as logical expression.

  $M(x)$: x is a human

  $G(y)$: y is shortcoming,

  $P(x, y)$: x has y.

  we can write it symbolically as

$$\forall x(M(x) \rightarrow \exists y(G(y) \wedge P(x, y)))$$

$$\forall x \exists y(M(x) \rightarrow (G(y) \wedge P(x, y)))$$

Negating Nested Quantifiers

- Statements involving nested quantifiers can be negated by successively applying the rules for negating statements involving a single quantifier.
- Example: Express the negation of the statement

  $\forall x \exists y (xy=1)$ so that no negation precedes a quantifiers.

$$\neg\forall x \exists y (xy=1)$$

$$\exists x \neg \exists y (xy=1)$$

$$\exists x \forall y \neg (xy=1)$$

$$\exists x \forall y (xy \neq 1)$$

Example

Use quantifiers and predicates express the fact that

$$\lim_{x \to a} f(x)$$

does not exist.

$$\neg \forall \epsilon > 0 \; \exists \delta > 0 \; \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

$$\equiv \exists \epsilon > 0 \; \neg \exists \delta > 0 \; \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

$$\equiv \exists \epsilon > 0 \; \forall \delta > 0 \; \neg \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

$$\equiv \exists \epsilon > 0 \; \forall \delta > 0 \; \exists x \; \neg (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

$$\equiv \exists \epsilon > 0 \; \forall \delta > 0 \; \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).$$

Exercises

- P.68: 12, 15, 19, 26, 30, 37, 45, 52

Formula and Normal Forms

- **Propositional normal form**
  - Conjunctive normal form
  - Disjunctive normal form

- Why it is needed to introduce propositional normal forms?

  Formulas can be transformed into standard forms
  - Make identification and comparison of two formulas
  - Become more convenient to determine whether a formula is tautology, contradiction or contingency
  - Become more convenient to find the assignments for which the formula is true or false
  - Can simplify the formula

- **Normal form for first order predicate logic**
  - Prenex normal forms

Propositional Formula

- Definition of wellformed formula (wff)

  (1) Each propositional variable is a formula, and the truth values T and F are formulas.

  (2) If A is formula, so is $(\neg A)$.

  (3) If A and B are formulas, so are $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, $(A \leftrightarrow B)$.

  (4) A string of symbols is a formula only as determined by (finitely many applications of) the above three rules.

Examples

- $\neg (p \vee q)$ √
- $p \rightarrow (q \rightarrow r)$ √
- $(p \wedge q) \leftrightarrow r$ √
- $pq \rightarrow r$ ×
- $\neg p \vee q) \rightarrow r$ ×

Propositional Normal Forms

- Formulas can be transformed into standard forms so that they become more convenient for symbolic manipulations and make identification and comparison of two formulas easier.
- There are two types of normal forms in propositional calculus: the disjunctive normal form(DNF) and the conjunctive normal form(CNF).
- DEFINITION: A literal is a variable or its negation.

Conjunctive Normal Form(CNF)

- DEFINITION: Disjunctions (conjunctions) with literals as disjuncts (conjuncts) are called disjunctive (conjunctive) clauses. Disjunctive and conjunctive clauses are simply called clauses.
- DEFINITION: A conjunction with disjunctive clauses as its conjunctions is said to be in conjunctive normal form.

$$(A_{11} \vee \ldots \vee A_{1n_1}) \wedge \ldots \wedge (A_{k1} \vee \ldots \vee A_{kn_K})$$

- Examples:
  - $p \wedge (q \vee r)$ √
  - $p \wedge \mathrm{F}$ √
  - $p \wedge ((p \wedge q) \vee r)$ ×

Disjunctive Normal Form(DNF)

- DEFINITION: A formula is said to be in disjunctive normal form if it is written as a disjunction, in which all the terms are conjunctions of literals.

$$(A_{11} \wedge \ldots \wedge A_{1n_1}) \vee \ldots \vee (A_{k1} \wedge \ldots \wedge A_{kn_K})$$

- Examples:
  - $(p \wedge q) \vee (p \wedge \neg q)$ √
  - $p \vee (q \wedge r)$ √
  - $\neg p \vee \mathrm{T}$ √
  - $\neg(p \wedge q) \vee r$ ×

More Examples

- p DNF & CNF
- $\neg p \vee q$ DNF & CNF
- $\neg p \wedge q \wedge \neg r$ DNF & CNF
- $\neg p \vee (q \wedge \neg r)$ DNF
- $\neg p \wedge (q \vee \neg r) \wedge (\neg q \vee r)$ CNF

How to obtain normal forms [1]

1) $A \rightarrow B \Leftrightarrow \neg A \vee B$
2) $A \leftrightarrow B \Leftrightarrow (\neg A \vee B) \wedge (A \vee \neg B)$
3) $A \leftrightarrow B \Leftrightarrow (A \wedge B) \vee (\neg A \wedge \neg B)$
4) $\neg\neg A \Leftrightarrow A$
5) $\neg(A_1 \wedge \ldots \wedge A_n) \Leftrightarrow \neg A_1 \vee \ldots \vee \neg A_n$
6) $\neg(A_1 \vee \ldots \vee A_n) \Leftrightarrow \neg A_1 \wedge \ldots \wedge \neg A_n$
7) $A \wedge (B_1 \vee \ldots \vee B_n) \Leftrightarrow (A \wedge B_1) \vee \ldots \vee (A \wedge B_n)$
   $(B_1 \vee \ldots \vee B_n) \wedge A \Leftrightarrow (A \wedge B_1) \vee \ldots \vee (A \wedge B_n)$
8) $A \vee (B_1 \wedge \ldots \wedge B_n) \Leftrightarrow (A \vee B_1) \wedge \ldots \wedge (A \vee B_n)$
   $(B_1 \wedge \ldots \wedge B_n) \vee A \Leftrightarrow (A \vee B_1) \wedge \ldots \wedge (A \vee B_n)$

How to obtain normal forms [2]

- By the replaceability of tautological equivalences, we can replace the preceding formulas on the left with the corresponding ones on the right to yield a formula tautologically equivalent to the original one.
- By (1)–(3) we eliminate $\rightarrow$ and $\leftrightarrow$.
- By (4)–(6) we eliminate $\neg$, $\wedge$, $\vee$, from the scope of $\neg$ such that any $\neg$ has only an atom as its scope.
- By (7) we eliminate $\vee$ from the scope of $\wedge$.
- By (8) we eliminate $\wedge$ from the scope of $\vee$.
- This method leads to obtaining the disjunctive or conjunctive normal forms.

Examples [1]

- Convert the following formula into a conjunctive normal form.

$$\neg((p \vee \neg q) \wedge \neg r)$$

$$\begin{aligned}
& \neg((p \vee \neg q) \wedge \neg r) \\
\Leftrightarrow\ & \neg(p \vee \neg q) \vee \neg\neg r && \text{De Morgan} \\
\Leftrightarrow\ & \neg(p \vee \neg q) \vee r && \text{Double negation} \\
\Leftrightarrow\ & (\neg p \wedge \neg\neg q) \vee r && \text{De Morgan} \\
\Leftrightarrow\ & (\neg p \wedge q) \vee r && \text{Double negation} \\
\Leftrightarrow\ & (\neg p \vee r) \wedge (q \vee r) && \text{Distributivity}
\end{aligned}$$

Examples [2]

- Convert the following formula into a conjunctive normal form.

$$(p_1 \wedge p_2) \vee (p_3 \wedge (p_4 \vee p_5))$$

$$(p_1 \vee p_3) \wedge (p_1 \vee p_4 \vee p_5) \wedge (p_2 \vee p_3) \wedge (p_2 \vee p_4 \vee p_5)$$

- Simplify the following conjunctive normal form.

$$(p \vee q) \wedge p \wedge (q \vee r) \wedge (p \vee \neg p \vee r) \wedge (\neg q \vee r)$$

$$(p \wedge r)$$

Examples [3]

- Convert the following formula into conjunctive and disjunctive normal forms. $\neg(p \vee q) \leftrightarrow (p \wedge q)$

solution:

Remark: Formulas (*) is in CNF. Both formulas (**) and (***) are in DNF.

$$\neg(p \vee q) \leftrightarrow (p \wedge q)$$
$$\equiv (\neg(p \vee q) \rightarrow (p \wedge q)) \wedge ((p \wedge q) \rightarrow \neg(p \vee q))$$
$$\equiv ((p \vee q) \vee (p \wedge q)) \wedge (\neg(p \wedge q) \vee \neg(p \vee q))$$
$$\equiv ((p \vee q \vee p) \wedge (p \vee q \vee q)) \wedge ((\neg p \vee \neg q) \vee (\neg p \wedge \neg q))$$
$$\equiv (p \vee q) \wedge (\neg p \vee \neg q \vee \neg p) \wedge (\neg p \vee \neg q \vee \neg q)$$
$$\equiv (p \vee q) \wedge (\neg p \vee \neg q)^{*}$$
$$\equiv ((p \vee q) \wedge \neg p) \vee ((p \vee q) \wedge \neg q)$$
$$\equiv (p \wedge \neg p) \vee (q \wedge \neg p) \vee (p \wedge \neg q) \vee (q \wedge \neg q)^{**}$$
$$\equiv (q \wedge \neg p) \vee (p \wedge \neg q)^{***}$$

Full disjunctive normal form

- Minterm & Maxterm
  - A *minterm* is a conjunctive of literals in which each variable is represented exactly once.
  - For example,
    - If a formula has the variables *p*, *q*, *r*, then $p \wedge \neg q \wedge r$ is a minterm, but $p \wedge \neg q$ and $p \wedge \neg p \wedge r$ are not.
  - Question:
    - If a formula has *n* variables, how many minterms are there? 2^n
- Full disjunctive normal form

  If a formula is expressed as a disjunction of minterms, it is said to be in *full disjunctive normal form*.

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

How to obtain full disjunctive normal form

- Any formula *A* is tautologically equivalent to a formula in full disjunctive normal form.
- First, obtain disjunctive normal form, then use of negation law and distributive laws to obtain full disjunctive forms

$$(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$$

$$(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$$

$$\equiv (p \wedge q \wedge (r \vee \neg r)) \vee (\neg p \wedge (q \vee \neg q) \wedge r) \vee ((p \vee \neg p) \wedge q \wedge r)$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r)$$

$$\equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$$

Disjunctive normal forms from truth tables

- So far we have shown how to find the truth table of a logical formula. The reverse is also possible. One can convert any given truth table into a formula. The formula obtained in this way is already in disjunctive normal form.
- Unfortunately, truth tables grow exponentially with the number of variables, which makes this method unattractive for formulas with many variables.

DNF from truth tables

- If a function, such as *f*, is given by truth table, we know exactly for which assignments it is true. Consequently, we can select the minterms that make the function true and form the disjunction of these minterms.

- Example: *f* a truth function, or a Boolean function with arguments *p, q, r* and range {0, 1}.

Example

p	q	r	f
T	T	T	T
T	T	F	F
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	T
F	F	F	F

The function f is true for three assignments:

1. p, q, r are all true.

2. p, $\neg q$, r are all true.

3. $\neg p$, $\neg q$, r are all true.

$f \Leftrightarrow (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$

Why ?

1) Each minterm is true for exactly one assignment.
2) A disjunction of minterms is true only if at least one of its constituents minterms is true.

CNF and Complementation

- If A is a formula containing only the connectives $\neg$, $\vee$ and $\wedge$, then its complement is formed by replacing all $\vee$ by $\wedge$, all $\wedge$ by $\vee$ and all atoms by their complements.
- Example: find the complement of the formula

$$A = (p \wedge q) \vee \neg r$$

$$(\neg p \vee \neg q) \wedge r$$

- Complementation can be used to obtain conjunctive normal forms from truth tables.

  One first determines the disjunctive normal form for $\neg f$. If the resulting disjunctive normal form is A, then $A \Leftrightarrow \neg f$, and the complement of A must be logically equivalent to f. (De Morgan's laws)

Examples

p	*q*	*r*	*f*
T	T	T	T
T	T	F	T
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	F
F	F	F	T

Solution: $\neg f$ is true for the following assignments:

p =T q =F r =T

p =T q =F r =F

p =F q =F r =T

The disjunctive normal form of $\neg f$ is therefore

$$(p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r)$$

This function has the complement

$$f \Leftrightarrow (\neg p \vee q \vee \neg r) \wedge (\neg p \vee q \vee r) \wedge (p \vee q \vee \neg r),$$

which is the desired conjunctive normal form.

Formula for first order predicate logic

- Atomic formula: A predicate name followed by a list of variables such as $P(x,y)$
- Wffs (well-formed formulas) are constructed by the following rules:

1) T and F are wffs.
2) Each propositional constant, and each propositional variable are wffs.
3) Each atomic formula is a wff.
4) If A, B, and C are wffs, then so are $\neg A$, $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, and $(A \leftrightarrow B)$.
5) If x is a variable, and A is a wff, then so are $\forall xA$ and $\exists xA$

Prenex Normal Forms

- DEFINITION: A formula is in prenex normal form if is of the form

$$Q_1 x_1 Q_2 x_2 \ldots Q_n x_n \, B$$

where Q_i (i =1, … , n) is ∀ or ∃ and the formula B is quantifier free.

- A formula with no quantifiers is regarded as a trivial case of a prenex normal form.
- Examples:
  - $\forall x P(x) \vee \forall x Q(x)$ ×
  - $\forall x \forall y \neg (P(x) \rightarrow Q(y))$ √
  - $\forall x \exists y R(x,y)$ √
  - $R(x,y)$ √
  - $\neg \forall x R(x,y)$ ×

Algorithm for prenex normal form [1]

- Any expression can be converted into prenex normal form. To do this, the following steps are needed:

  1. Eliminate all occurrences of $\rightarrow$ and $\leftrightarrow$ from the formula in question.

  2. Move all negations inward such that, in the end, negation only appear as part of literals.

  3. Standardize the variables apart (when necessary).

  4. The prenex normal form can now be obtained by moving all quantifiers to the front of the formula.

Step 1/2

- Step 1:
  - $A \rightarrow B \Leftrightarrow \neg A \vee B$
  - $A \leftrightarrow B \Leftrightarrow (\neg A \vee B) \wedge (A \vee \neg B)$
  - $A \leftrightarrow B \Leftrightarrow (A \wedge B) \vee (\neg A \wedge \neg B)$
- Step 2:
  - $\neg\neg A \Leftrightarrow A$
  - $\neg\exists x A(x) \Leftrightarrow \forall x \neg A(x)$
  - $\neg\forall x A(x) \Leftrightarrow \exists x \neg A(x)$

Step 3 (standardize variable apart)

- Theorem (Replaceability of bound variable symbols): Suppose A results from A' by replacing in A some (not necessarily all) occurrences of $Q(x)B(x)$ by $Q(y)B(y)$. Then $A \Leftrightarrow A'$.
- Example: Standardize all variables apart in the following formula

$$\forall x(P(x)\rightarrow Q(x)) \wedge \exists x Q(x) \wedge \exists z P(z) \wedge \exists z(Q(z)\rightarrow R(x))$$

Solution: Use y for x in $\forall x$, u for x in $\exists x Q(x)$, and w for z in $\exists z P(z)$ to obtain:

$$\forall y(P(y)\rightarrow Q(y)) \wedge \exists u Q(u) \wedge \exists w P(w) \wedge \exists z(Q(z)\rightarrow R(x))$$

Step 4 [1]

- $A \wedge \exists x B(x) \Leftrightarrow \exists x\,(A \wedge B(x))$
- $A \wedge \forall x B(x) \Leftrightarrow \exists x\,(A \wedge B(x))$
- $A \vee \exists x B(x) \Leftrightarrow \exists x\,(A \vee B(x))$
- $A \vee \forall x B(x) \Leftrightarrow \forall x\,(A \vee B(x))$
- These equivalences essentially show that if a formula A has a truth value that does not depend on x, then one is allowed to quantify over x.

Step 4 [2]

- $\forall x A(x) \wedge \forall x B(x) \Leftrightarrow \forall x\,(A(x) \wedge B(x))$
- $\exists x A(x) \vee \exists x B(x) \Leftrightarrow \exists x\,(A(x) \vee B(x))$

$\forall x A(x) \vee \forall x B(x) \Leftrightarrow \forall x\,(A(x) \vee B(x))$ X

$\exists x A(x) \wedge \exists x B(x) \Leftrightarrow \exists x\,(A(x) \wedge B(x))$ X

- $\forall x \forall y\, A(x,y) \Leftrightarrow \forall y \forall x\, A(x,y)$
- $\exists x \exists y\, A(x,y) \Leftrightarrow \exists y \exists x\, A(x,y)$

- $Q_1 x A(x) \wedge Q_2 y B(y) \Leftrightarrow Q_1 x Q_2 y (A(x) \wedge B(y))$
- $Q_1 x A(x) \vee Q_2 y B(y) \Leftrightarrow Q_1 x Q_2 y (A(x) \vee B(y))$

For example $\forall x A(x) \wedge \exists y B(y) \Leftrightarrow \forall x \exists y\,(A(x) \wedge B(y))$

Example

- Find the prenex normal form of

$$\forall x((\exists y R(x,y) \wedge \forall y \neg S(x,y)) \rightarrow \neg(\exists y R(x,y) \wedge P))$$

According to Step 1, we must eliminate $\rightarrow$, which yields

$$\forall x(\neg(\exists y R(x,y) \wedge \forall y \neg S(x,y)) \vee \neg(\exists y R(x,y) \wedge P))$$

We move all negations inwards, which yields:

$$\forall x(\forall y \neg R(x,y) \vee \exists y S(x,y) \vee \forall y \neg R(x,y) \vee \neg P)$$

Next, all variables are standardized apart:

$$\forall x(\forall y_1 \neg R(x,y_1) \vee \exists y_2 S(x,y_2) \vee \forall y_3 \neg R(x,y_3) \vee \neg P)$$

We can now move all quantifiers in front, which yields

$$\forall x \forall y_1 \exists y_2 \forall y_3 (\neg R(x,y_1) \vee S(x,y_2) \vee \neg R(x,y_3) \vee \neg P)$$

Prenex DNF and CNF

- Step1: Prenex normal form
- Step 2: Prenex DNF or CNF

Exercises [1]

1. Give the simplest DNF and CNF of the following formulas:

   1) $((P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)) \wedge R$

   2) $P \vee (\neg P \vee (Q \wedge \neg Q))$

   3) $(P \wedge (Q \wedge S)) \vee (\neg P \wedge (Q \wedge S))$

2. Give the full DNF of the following formulas:

   1) $(\neg P \vee \neg Q) \rightarrow (P \leftrightarrow \neg Q)$

   2) $(\neg R \wedge (Q \rightarrow P)) \rightarrow (P \rightarrow (Q \vee R))$

   3) $(P \rightarrow (Q \wedge R)) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))$

Exercises [2]

3. Give the prenex normal forms of the following formulas:

   1) $(\forall x)(P(x) \rightarrow (\exists y)Q(x,y))$

   2) $(\forall x)(\forall y)(((\exists z)P(x,y,z) \wedge (\exists u)Q(x,u)) \rightarrow (\exists v)Q(y,v))$

4. Give the prenex DNF and CNF of the following formulas:

   1) $((\exists x)P(x) \vee (\exists x)Q(x)) \rightarrow (\exists x)(P(x) \vee Q(x))$

   2) $(\forall x)(P(x) \rightarrow Q(x,y)) \rightarrow ((\exists y)\mathrm{P}(y) \wedge (\exists z)Q(y,z))$

Chapter 1

- 1.1-1.2 Propositional Logic
- 1.3 Propositional Equivalences
- 1.4 Predicates and Quantifiers
- 1.5 Nested Quantifiers
- 1.6 Rules of Inference
- 1.7 Introduction to Proofs
- 1.8 Proof Methods and Strategy

1.6 Rules of Inference

Two important questions that arise in the study of mathematics:

- When is a mathematical argument correct?
- What methods can be used to construct mathematical arguments?

Introduction

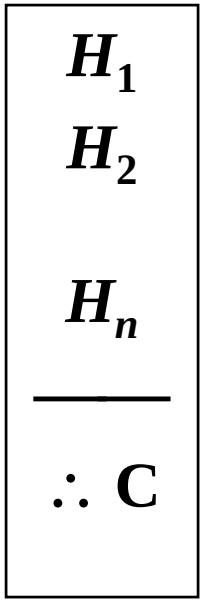
- Proofs in mathematics are valid arguments that establish the truth of mathematical statements.
- By an argument, we mean a sequence of statements that end with a conclusion.
- By valid, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premises. That is , an argument is valid iff it is impossible for all the premises to be true and the conclusion to be false.
- Fallacies: Some common forms of incorrect reasoning which lead to invalid arguments.

Valid Arguments in Propositional Logic

- Definition: An argument in propositional logic is a sequence of propositions. All but the final proposition in the argument are called premises and the final proposition is called the conclusion. An argument is valid if the truth of all its premises implies that the conclusion is true.
- An argument form in propositional logic is a sequence of compound propositions involving propositional variables.

Rules of Inference for Propositional Logic

- Rule of inference: an implication that is a tautology which is then used to draw conclusion from known assertions.



$H_1 \wedge H_2 \wedge \ldots \wedge H_n \rightarrow C$ **is tautology**

$H_1, H_2, \ldots H_n$ are the hypotheses

C is the conclusion.

"$\therefore$" means 'therefore' or 'it follows that'

How to show that an argument is valid

- using truth table
- First, establish the validity of some relatively simple argument forms, called *rules of inference*. Then, use these rules of inference as building blocks to construct more complicated valid argument forms.

Rules of Inference for Propositional Logic

- Addition
- Simplification
- Conjunction
- Modus ponens
- Modus Tollens
- Hypothetical syllogism
- Disjunctive syllogism
- Constructive dilemma
- Resolution

Important rules [1]

$\dfrac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition 附加
$\dfrac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification 化简
$\dfrac{\begin{array}{c}p\\q\end{array}}{\therefore p \wedge q}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction 合取
$\dfrac{\begin{array}{c}p\\p \rightarrow q\end{array}}{\therefore q}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus Ponens 假言推理

Important rules [2]

$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus Tollens 拒取式
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical Syllogism 假言三段论
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive Syllogism 析取三段论

Important rules [3]

$p \to r$ $q \to r$ $p \vee q$ $\therefore r$	$((p \to r) \wedge (q \to r) \wedge (p \vee q)) \to r$	Constructive Dilemma 构造两难
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution 消解

Using Rules of Inference to Build Arguments

- An argument is valid if
  - whenever all hypotheses are true, the conclusion is also true
- To prove that an argument is valid:
  - Assume the hypotheses are true
  - Use the rules of inference and logical equivalences to determine that the conclusion is true

Example [1]

- Consider the following logical argument:
  - If horses fly or cows eat artichokes, then the mosquito is the national bird.
  - If the mosquito is the national bird then peanut butter tastes good on hot dogs.
  - But peanut butter tastes terrible on hot dogs.
  - Therefore, cows don’t eat artichokes.

Example [1]

Assign propositional variables to each component in the argument:

p	**Horses fly**
q	**Cows eat artichokes**
r	**The mosquito is the national bird**
s	**Peanut butter tastes good on hot dogs**

Represent the argument using the variables

$(p \vee q) \rightarrow r$

$r \rightarrow s$

$\neg s$

$\therefore \neg q$ **Conclusion**

Example [1]

1. $(p \vee q) \rightarrow r$ **Hypothesis**
2. $r \rightarrow s$ **Hypothesis**
3. $(p \vee q) \rightarrow s$ **Hypothetical syllogism on 1. and 2.**
4. $\neg s$ **Hypothesis**
5. $\neg(p \vee q)$ **Modus tollens on 3. and 4.**
6. $\neg p \wedge \neg q$ **DeMorgan on 5.**
7. $\neg q \wedge \neg p$ **Commutative on 6.**
8. $\neg q$ **Simplification on 7.**

We got our conclusion of *"cows don't eat artichokes"*

Example [2]

- Example 6 (page 77)
- It is not sunny this afternoon (p)
- And it is colder than yesterday (q)
- We will go swimming (r) only if it is sunny
- If we do not go swimming, then we will take a canoe trip (s)
- If we take a canoe trip, then we will be home by sunset (t)
- Conclusion: We will be home by sunset.

Example [2]

1.	$\neg p \vee q$	Hypothesis
2.	$\neg p$	Simplification on 1
3.	$r \rightarrow p$	Hypothesis
4.	$\neg r$	Modus tollens on 2 and 3
5.	$\neg r \rightarrow s$	Hypothesis
6.	s	Modus tollens on 4 and 5
7.	$s \rightarrow t$	Hypothesis
8.	t	

We got our conclusion of *"we will be home by sunset"*

Example [3]

- Show that $\neg w$ logically follows from the hypotheses

$$(w \vee r) \rightarrow v, v \rightarrow (c \vee s), s \rightarrow u, \neg c \wedge \neg u$$

Step	Reason
1. $(w \vee r) \rightarrow v$	Hypothesis
2. $v \rightarrow (c \vee s)$	Hypothesis
3. $(w \vee r) \rightarrow (c \vee s)$	Hypothetical syllogism using (1) and (2)
4. $\neg c \wedge \neg u$	Hypothesis
5. $\neg u$	Simplification using (4)
6. $s \rightarrow u$	Hypothesis
7. $\neg s$	Modus tollens using (5) and (6)
8. $\neg c$	Simplification using (4)
9. $\neg s \wedge \neg c$	Conjunction using (7) and (8)
10. $\neg (c \vee s)$	(9) and De morgan
11. $\neg (w \vee r)$	Modus tollens using (3) and (10)
12. $\neg w \wedge \neg r$	(10) and De morgan
13. $\neg w$	Simplification using (12)

Notes

- If the conclusion is given in form $p \rightarrow q$, we can convert the original problem to

$$p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge p \Rightarrow q$$

- This method is based on

$$(p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge p) \rightarrow q \Leftrightarrow (p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow (p \rightarrow q)$$

Resolution

- If $p \vee q$ and $\sim p \vee r$ are both true, then $q \vee r$ is true.
- In a proof by resolution, the hypotheses and the conclusion are written as clause.
- A clause consists of terms separated by or's, where each term is a variable or the negation of a variable.

Example

- (P.78: example 9) Show that the hypotheses $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$

Solution:

$$(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$$

$$r \rightarrow s \equiv \neg r \vee s$$

$$((p \vee r) \wedge (\neg r \vee s)) \rightarrow (p \vee s)$$

Fallacies

- Avoid: An implication that is a contingency that is often incorrectly used to draw conclusions, which is not a tautology.
- Types of the fallacies:
  - fallacy of affirming the conclusion

    $[(p \rightarrow q) \wedge q] \rightarrow p$ is not a tautology **(p=F, q=T)**
  - fallacy of denying the hypothesis

    $[(p \rightarrow q) \wedge \neg p] \rightarrow \neg q$ is not a tautology **(p=F, q=T)**
  - circular reasoning

Examples

- If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics. Therefore, you did every problem in this book. X
- If you do every problem in this book, then you will learn discrete mathematics. You did not do every problem in this book. Therefore, you did not learn discrete mathematics. X
- One or more steps of the proof are based upon the truth of the statement being proved. X

Rules of inference for quantified statements

$\forall x P(x)$ $\therefore P(c)$ if $c \in U$	Universal instantiation
$P(c)$ for an arbitrary $c \in U$ $\therefore \forall x P(x)$	Universal generalization
$\exists x P(x)$ $\therefore P(c)$ for some element $c \in U$	Existential instantiation
$P(c)$ for some element $c \in U$ $\therefore \exists x P(x)$	Existential generalization

Example[1]

- Everyone in the discrete math class has taken a CS course. Marla is a student in the discrete class. Therefore, Marla has taken a CS course.

$D(x)$: x is in the discrete math class
$C(x)$: x has taken a CS course

$$\left.\begin{array}{l} \forall x(D(x) \rightarrow C(x)) \\ D(\text{Marla}) \end{array}\right\} \text{Premises}$$
$$\therefore\ C(\text{Marla})$$

1. $\forall x(D(x) \rightarrow C(x))$ Premise
2. $D(\text{Marla}) \rightarrow C(Marla)$ Univ. Inst. using 1.
3. $D(\text{Marla})$ Premise
4. $C(\text{Marla})$ Modus ponens using 2. and

Example[2]

- Consider the following logical argument, is it valid?

  Each member of the committee is teacher and expert. Some members of the committee are younger. Therefore, some members of the committee are young experts.

Let *F*(*x*) be "*x* is a member of the committee", let *G*(*x*) be "*x* is an expert",

let *H*(*x*) be "*x* is a teacher"

let *R*(*x*) be " *x* is a younger."

Then the premises is $\forall x(F(x) \rightarrow G(x) \wedge H(x)) \wedge \exists x(F(x) \wedge R(x))$

The conclusion is $\exists x(F(x) \wedge G(x) \wedge R(x))$

Combining Rules of Inference for Propositions and Quantified Statements (Proof of Socrates assertion)

- All human beings are mortal p
- Socrates is a human being q
- Therefore Socrates is mortal r
- $p \wedge q \rightarrow r$ *is not a tautology*, therefore we can't prove this conclusion by propositional logic. But we know by intuition that Socrates assertion is correct. We must look for another way of proof.

$$\forall x(P(x) \rightarrow Q(x))$$

$P(a)$, where a is a particular element in the domain

$$\therefore Q(a)$$

Universal Modus Ponens

H(x): x is a human being;
M(x):x is mortal;
S: Scorates
Socrates assertion can be interpreted as following
$\forall x(H(x) \rightarrow M(x))$
$H(S)$
$\therefore C(S)$

1. $\forall x(H(x) \rightarrow M(x))$	Premise
2. $H(S) \rightarrow M(S)$	Univ. Inst. using 1.
3. $H(S)$	Premise
4. $M(S)$	Modus ponens using 2. And 3

Exercises

- P.82: 19, 24, 29, 33, 35

1.7 Introduction to Proofs

- A *theorem* is a valid logical assertion which can be proved using
  - *Axioms*: statements which are given to be true
  - *Proposition*: Less important theorem
  - *Proof* : A valid argument that establishes the truth of a theorem
- A *lemma* is a ‘pre-theorem’ or a result which is needed to prove a theorem.
- A *corollary* is a ‘post-theorem’ or a result which follows directly from a theorem.
- *A conjecture* is a statement whose truth value is unknown

Methods of proving theorems

- Direct proof
- Indirect proof
- Proof by contradiction

Proof Basics

- We want to establish the truth of $p \rightarrow q$
- p may be a conjunction of other hypotheses
- $p \rightarrow q$ is a *conjecture* until a proof is produced

To prove a theorem of the form $\forall x\,(P(x) \rightarrow Q(x))$, we can show that $P(c) \rightarrow Q(c)$ is true, where c is an arbitrary element of the domain, and then apply universal generalization.

Direct Proof

- Assume the hypotheses are true
- Use rules of inference and any logical equivalences to establish the truth of the conclusion

- Example: The proof we did earlier about cows not eating artichokes was an example of a direct proof

If n is odd, then n^2 is odd.

Proof:

Assume that the hypothesis of this implication is true, namely, suppose that n is odd.

Then $n = 2k + 1$, where k is an integer.

It follows that

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Therefore,

n^2 is odd (it is 1 more than twice an integer).

Indirect Proof (Proof by Contraposition)

- A direct proof of the contrapositive
  - Proof of $(\neg q) \rightarrow (\neg p)$
  - Assume that $\neg q$ is true i.e., q is false
  - Use rules of inference and logical equivalences to show that $\neg p$ is true i.e., p is false
- Note: In order to show that a conjunction of hypotheses is false is suffices to show just one of the hypotheses is false.

- P.78, Example 3: If $3n+2$ is odd, then n is odd.

Proof by Contradiction [1]

- *Goal:* To establish the truth of the 'theorem' p.
- Assumes the conclusion p is false. Find a contradiction $(q \wedge \neg q)$ such that

$$\neg p \rightarrow \mathrm{F} \Leftrightarrow \mathrm{T}$$

- To prove that p is true, you have to show that $\neg p$ leads to a contradiction i.e., you have to prove that $\neg p \rightarrow \mathrm{F}$ is true.
- P.90, Example 11: $\sqrt{2}$ is irrational.

Prove: There is no largest prime number

- Let *p* by the proposition 'there is no largest prime number' .
- Suppose that ¬*p* is true, namely, there is a largest prime number, denoted by *s*.
- Hence, the set of all primes lie between 1 and *s*.
- Form the product of these primes:

  $r = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot \ldots \cdot s$.
- But *r* + 1 is a prime larger than *s*. Why?
- This is a contradiction since we have shown that ¬*p* implies both *q* and ¬*q* where *q* is the statement that *s* is the largest prime number.
- Hence, ¬*p* is false, so that *p*: 'there is no largest prime number' is true.

Proof by Contradiction [2]

- An indirect proof of an implication $p \rightarrow q$ can be rewritten as a proof by contradiction.
- Assume that both p and $\neg q$ are true.
- Then use a direct proof to show that

$$\neg q \rightarrow \neg p$$

- This leads to the contradiction $p \wedge \neg p$.

- P.91, Example 12: If $3n+2$ is odd, then n is odd.

Vacuous Proof

- If we know one of the hypotheses in p is false then $p \rightarrow q$ is *vacuously* true.
- F $\rightarrow$ T and F $\rightarrow$ F are both true.

- Example:
  - If I am both rich and poor, then hurricane Fran was a mild breeze.
  - The hypotheses $(p \wedge \neg p)$ form a contradiction, therefore q follows from the hypotheses vacuously.
- Example:
  - $P(0)$ is true where $P(n)$: If $n > 1$, then $n^2 > n$.

Trivial Proof

- If we know q is true, then $p \rightarrow q$ is true
- F $\rightarrow$ T and T $\rightarrow$ T are both true.

- Example:
  - If it's raining today then the empty set is a subset of every set.
  - The assertion is *trivially* true independent of the truth value of ***p***.
- Example:
  - ***P*(0)** is true where ***P*(*n*):** If $\boldsymbol{a \geq b > 0}$, then $\boldsymbol{a^n \geq b^n}$.

Proving an equivalence

- To prove that $p \Leftrightarrow q$, you need to show that $p \leftrightarrow q$ is a tautology.
- You can do that by showing that $p \rightarrow q$ and $q \rightarrow p$ are both true since,

$$p \leftrightarrow q \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)]$$

- Example: The integer n is odd if and only if n^2 is odd.

Mistakes in Proof

- P.93-95: Self-study

Exercises

- P.95: 7, 17, 33, 38

1.8 Proof Methods and Strategy

- Proof by cases
- Existence proof

Exhaustive Proof and Proof by Cases

- Break the premise of $p \rightarrow q$ into an equivalent disjunction of the form $p_1 \vee p_2 \vee \ldots \vee p_n$
- Then use the tautology

$$((p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \ldots \wedge (p_n \rightarrow q))$$
$$\leftrightarrow ((p_1 \vee p_2 \vee \ldots \vee p_n) \rightarrow q)$$

- Each of the implications $p_i \rightarrow q$ is a *case.*
- You must
  - Convince the reader that the cases are inclusive (i.e., they exhaust all possibilities)
  - Establish all implications

Proof by Cases [2]

- Example: If n is an integer not divisible by 3, then $n^2 \equiv 1 \pmod 3$.

  p: n is an integer not divisible by 3

  q: $n^2 \equiv 1 \pmod 3$

- $p \Leftrightarrow (p_1 \vee p_2)$

  where, p_1: $n \equiv 1 \pmod 3$

  p_2: $n \equiv 2 \pmod 3$

- Now prove that $((p_1 \rightarrow q) \wedge (p_2 \rightarrow q))$ is true i.e., both the cases $(p_1 \rightarrow q)$ and $(p_2 \rightarrow q)$ are true.

Leveraging proof by cases

- Example: Prove that if n is an integer not divisible by 2 or 3, then n^2-1 is divisible by 24.

  $n = 6k+1$

  $n = 6k+5$

- Example: Show that there are no solutions in integers x and y of $x^2+3y^2=8$.

  $x = -2,-1,0,1,2$

  $y = -1,0,1$

Exhaustive Proof

- An exhaustive proof is a special type of proof by cases where each case involves checking a single example.
- P.100, Example 8

Existence Proof

- The proof of $\exists x P(x)$ is called an *existence proof*.
  - Constructive
  - Non-constructive
- Constructive existence proof
  - Find an element c in the universe of discourse such that $P(c)$ is true
- Non-constructive existence proof
  - Assume no c exists that makes $P(c)$ true
  - Derive a contradiction

Example (Constructive existence proof)

- Show that there are n consecutive composite positive integers for every positive integer n.

Proof:

- Let $x = (n + 1)! + 1$.
- Consider the integers $x + 1, x + 2, \ldots, x + n$.
- Note that $i + 1$ divides $x + i = (n + 1)! + (i + 1)$ for $i = 1, 2, \ldots, n$.
- Hence, n consecutive composite positive integers have been given.
- Note that in the solution a number x such that $x + i$ is composite for $i = 1, 2, \ldots, n$ has been produced.
- Hence, this is an example of constructive existence proof.

Example (Non-constructive existence proof)

- Show that there exist irrational numbers x and y such that x^y is rational.

Solution: By Example 11 in Section 1.7 we know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$. If it is rational, we have two irrational numbers x and y with x^y rational, namely, $x = \sqrt{2}$ and $y = \sqrt{2}$. On the other hand if $\sqrt{2}^{\sqrt{2}}$ is irrational, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ so that $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2}\cdot\sqrt{2})} = \sqrt{2}^2 = 2$.

This proof is an example of a nonconstructive existence proof because we have not found irrational numbers x and y such that x^y is rational. Rather, we have shown that either the pair $x = \sqrt{2}, y = \sqrt{2}$ or the pair $x = \sqrt{2}^{\sqrt{2}}, y = \sqrt{2}$ have the desired property, but we do not know which of these two pairs works! ◀

Uniqueness Proofs

Two steps:

- Existence: We show that an element x with the desired property exists.
- Uniqueness: We show that if $y \neq x$, then y does not have the desired property.
- P.103: Example 13

Proof Strategies

- Direct proof
- Proof by contraposition
- Proof by contradiction

- … …

Forward and backward reasoning

- P.105, Example 14: Given two positive real numbers a and b, their arithmetic mean is $(a+b)/2$ and their geometric mean is $(ab)^{1/2}$, the arithmetic mean is always greater than the geometric mean.
- P.106, Example 15: Suppose that two people play a game taking turns removing 1, 2, or 3 stones at a time from a pile that begins with 15 stones. The person who removes the last stone wins the game. Show that the first player can win the game no matter what the second player does.

Adapting existing proofs

- P.106, Example 16: We have proved that $\sqrt{2}$ is irrational. We now conjecture that $\sqrt{3}$ is irrational. Can we adapt the proof of $\sqrt{2}$ being irrational to show that $\sqrt{3}$ is irrational?

- A good tip is to look for existing proofs that you might adapt when you are confronted with proving a new theorem, particularly when the new theorem seems similar to one you already proved.

Looking for Counterexamples

Goal: To establish that $\neg\forall x P(x)$ is true

(or $\neg\forall x P(x)$ is false).

Recall: $\neg\forall x P(x) \Leftrightarrow \exists x \neg P(x)$

- $f(n)=n^2-n+41$ is prime for all positive integers n.

  $f(41)=41^2-41+41$

Proof Strategy in Action

- The discovery process in mathematics:

  Begin with exploring concepts and examples, asking questions, formulating conjectures, and attempting to settle these conjecture either by proof or by counterexample.

Self-learning

Additional Proof Methods

- Mathmatical induction
- Structural induction
- The Cantor diagonalization method
- Combinatorial proofs

Exercises

- P.113: 13, 15, 24